

Replication Report: Field-Free Deterministic SOT Switching via Cluster-Octupole Out-of-Plane Spin Polarization σ_z (You et al., 2021)

Automated replication (macrospin LLG)

July 20, 2026

1 Target claim

You *et al.* (2021, Mn_3SnN antiperovskite noncollinear AFM) report that an out-of-plane spin polarization $\sigma_z \propto \mathbf{H}_{so} \times \mathbf{T}$, generated by the cluster magnetic octupole $\mathbf{T}$ when the current $\mathbf{J} \parallel \mathbf{T}$, drives **field-free deterministic** spin-orbit-torque (SOT) switching of an adjacent perpendicular ferromagnet $(\text{Co/Pd})_3$. The switching polarity is set by the current sign, and σ_z (hence the deterministic field-free switching) *vanishes* when $\mathbf{J} \perp \mathbf{T}$. Measured torque ratios: $\theta_{AD,z} = 0.003$, $\theta_{FL,z} = 0.053$; $J_{crit} \approx 9 \times 10^6 \text{ A cm}^{-2}$.

2 Nature of the paper and scope of replication

The paper is predominantly *experimental* (epitaxial growth, XRD, SQUID, AHE, ST-FMR, MOKE, transport). No simulation workflow or code is provided. However, its central mechanistic assertion is a *symmetry-defined theory result*: the antidamping σ_z torque $\tau_C = \mathbf{m} \times (\mathbf{m} \times \sigma_z)$ lifts the $\pm\hat{z}$ degeneracy of a PMA magnet with no external field. We reproduce this from scratch with a single-macrospin Landau–Lifshitz–Gilbert (LLG) integration. The materials-growth, ST-FMR fitting, and MOKE imaging portions are *not* reproducible without the physical samples and instruments; hence the overall verdict is **PARTIAL**.

3 Model

Unit magnetization $\mathbf{m}$, uniaxial PMA along $\hat{z}$ ($\mathbf{H} = h_k m_z \hat{z}$, $h_k = 1$), Gilbert damping $\alpha = 0.1$, reduced units, **zero external field everywhere**. RK4 integration of

$$\frac{d\mathbf{m}}{dt} = -\mathbf{m} \times \mathbf{H} - \alpha \mathbf{m} \times (\mathbf{m} \times \mathbf{H}) + \tau_{DL} \mathbf{m} \times (\mathbf{m} \times \mathbf{p}) + \tau_{FL} \mathbf{m} \times \mathbf{p},$$

with SOT amplitudes scaled from the paper’s ratios, $\tau_{DL} = C\theta_{AD,z}$, $\tau_{FL} = C\theta_{FL,z}$, current pulse ON for 60% of the run then relaxed.

- **Case A** ($\mathbf{J} \parallel \mathbf{T}$, σ_z present): polarization $\mathbf{p} \approx \hat{z}$ (dominant out-of-plane, small σ_y).
- **Case B** ($\mathbf{J} \perp \mathbf{T}$, σ_z absent): $\mathbf{p} = \hat{y}$ (conventional in-plane σ_y only).

Determinism test: initialise both $m_z = +1$ and $m_z = -1$; require the final state to depend *only* on current sign, not on initial state.

4 Results

Case	init	final (+ I)	final ($-I$)
A (σ_z present)	up	down	up
A (σ_z present)	down	down	up
B (σ_z absent)	up	up	up
B (σ_z absent)	down	down	down

Table 1: Case A: final state set by current sign, independent of initial state $\Rightarrow$ deterministic field-free switching. Case B: final state tracks initial state, no switching. Threshold current $C_{crit} \approx 12$ (reduced units).

Case A reproduces **deterministic field-free switching with current-sign-controlled polarity**. Case B shows **no** deterministic field-free switching. This matches the paper’s key control experiment (Fig. 4c vs. 4f: switching along $[001]$ $J \parallel T$, no switching along $[110]$ $J \perp T$).

5 Verdict: PARTIAL

Coverage 6/10 — the core mechanistic theory claim is fully reproduced; the experimental growth/ST-FMR/MOKE bulk of the paper is not reproducible here. **Agreement 8/10** — qualitative and structural agreement is complete (deterministic switching present iff σ_z present; polarity set by current sign; finite threshold current). Absolute J_{crit} in physical units and the small $\theta_{AD,z}$ are not quantitatively benchmarked, only used as the FL/AD ratio.

6 Physics summary

The antidamping component of the out-of-plane spin polarization, $\tau_C \propto \mathbf{m} \times (\mathbf{m} \times \boldsymbol{\sigma}_z)$, is the only torque term that can deterministically break the up/down symmetry of a perpendicular magnet without an external field. Our macrospin LLG confirms this: with σ_z present the final magnetization is fixed by the current sign regardless of the starting state, and with σ_z removed (only σ_y) the magnet is not switched — exactly the $J \parallel T$ vs. $J \perp T$ dichotomy the paper attributes to the cluster magnetic octupole.